Q proves distinct numerals inequal

The Open Logic Project contributors

openlogic:incompleteness/representability-in-q:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-incompleteness-representability-in-q/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: both cases are complete; the quantifier-axiom instantiations are written out by the author

Editorial note. The label lem:q-proves-neq names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Q axioms for successor and zero. Q_2: 0 != x'; the axiom forall x (0 != x'); and the injectivity axiom forall x forall y (x' = y' -> x = y), plus quantifier axiom instantiation.. Both cases discharge their contradiction or contraposition through these axioms instantiated at numerals.

Lemma 0.1. *Given natural numbers n and m, if $n \neq m$, then* $\mathrm{Th}Q \vdash \overline{n} \neq \overline{m}$.

Proof. Use induction on n to show that for every m, if $n \neq m$, then $Q \vdash \overline{n} \neq \overline{m}$.

In the base case, $n = 0$. If m is not equal to 0, then $m = k + 1$ for some natural number k. We have an axiom that says $\forall x\, 0 \neq x'$. By a quantifier axiom, replacing x by $\overline{k}$, we can conclude $0 \neq \overline{k}'$. But $\overline{k}'$ is just $\overline{m}$.

In the induction step, we can assume the claim is true for n, and consider $n + 1$. Let m be any natural number. There are two possibilities: either $m = 0$ or for some k we have $m = k + 1$. The first case is handled as above. In the second case, suppose $n + 1 \neq k + 1$. Then $n \neq k$. By the induction hypothesis for n we have $\mathrm{Th}Q \vdash \overline{n} \neq \overline{k}$. We have an axiom that says $\forall x\, \forall y\, x' = y' \to x = y$. Using a quantifier axiom, we have $\overline{n}' = \overline{k}' \to \overline{n} = \overline{k}$. Using propositional logic, we can conclude, in $\mathrm{Th}Q$, $\overline{n} \neq \overline{k} \to \overline{n}' \neq \overline{k}'$. Using modus ponens, we can conclude $\overline{n}' \neq \overline{k}'$, which is what we want, since $\overline{k}'$ is $\overline{m}$. □